Sound

The big idea

Sound is a form of energy produced by **vibrating** objects. It needs a **medium** (solid, liquid or gas) to travel, which is why there is no sound in space.

How sound travels

A vibrating object pushes and pulls the particles of the medium, creating regions of high pressure (**compressions**) and low pressure (**rarefactions**). The particles move back and forth **along** the direction of travel, so sound is a **longitudinal wave**. (In transverse waves, like ripples on water, particles move perpendicular to the wave.)

Characteristics of a sound wave

Term	Meaning
Wavelength (λ)	Distance between two consecutive compressions (unit m)
Frequency (ν)	Number of oscillations per second (hertz, Hz)
Time period (T)	Time for one oscillation; T = 1/ν
Amplitude	Maximum disturbance; decides **loudness**
Pitch	How high or low; decided by **frequency**
Quality (timbre)	Distinguishes sounds of the same pitch and loudness

WAVE SPEED

$$v = \lambda \times \nu$$

A SOUND WAVE HAS FREQUENCY 2 KHZ AND WAVELENGTH 35 CM. FIND ITS SPEED.

$v = 0.35 \times 2000 =$ **700 m/s**.

The speed of sound depends on the medium and temperature: about **346 m/s in air at 25 °C**, faster in water and fastest in solids (≈5000 m/s in steel). Light is much faster, which is why we see lightning before hearing thunder.

Reflection of sound

Sound reflects like light. An **echo** is a reflected sound heard separately. Because our ear keeps a sound for about **0.1 s**, the reflecting surface must be at least about **17.2 m** away (in air at 22 °C).

Reverberation is the persistence of sound from repeated reflections; auditoriums reduce it with sound-absorbing panels, curtains and seat fabric. Megaphones, stethoscopes and curved ceilings use repeated reflection.

Range of hearing

Humans hear **20 Hz to 20,000 Hz**. Below 20 Hz is **infrasound** (elephants, whales, earthquakes); above 20 kHz is **ultrasound** (dogs, bats, dolphins).

USES OF ULTRASOUND

Cleaning hard-to-reach parts, detecting cracks in metal, **echocardiography** (heart), **ultrasonography** (internal organs, foetus), breaking kidney stones, and **SONAR** (finding ocean depth and underwater objects).

SONAR

A ship's ultrasound pulse returns from the sea-bed after 2 s. Speed in water is 1500 m/s.
Total distance = 1500 × 2 = 3000 m, so depth = **1500 m** (half, because the sound goes down and back).

Quick recap

- Sound is a longitudinal wave of compressions and rarefactions; needs a medium.
- $v = \lambda\nu$; frequency → pitch, amplitude → loudness.
- Echo needs at least 17.2 m; ultrasound has many uses including SONAR.

Practice questions

1. Sound cannot travel through
   (a) water (b) steel (c) air (d) vacuum
2. The pitch of a sound depends on its
   (a) amplitude (b) frequency (c) speed (d) wavelength only in solids
3. The audible range of human hearing is
   (a) 2 Hz to 2000 Hz (b) 20 Hz to 20,000 Hz (c) 200 Hz to 200,000 Hz (d) 20 kHz to 200 kHz
4. Sound waves in air are
   (a) longitudinal (b) transverse (c) electromagnetic (d) both transverse and longitudinal
5. A wave has frequency 500 Hz and wavelength 0.7 m. Its speed is
   (a) 71.4 m/s (b) 350 m/s (c) 700 m/s (d) 500.7 m/s

Answers: 1. d 2. b 3. b 4. a 5. b